

EVENTFORCE MANAGEMENT SYSTEM

SALESFORCE IMPLEMENTATION

Project Documentation

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USER STORY

The EventForce Management System is designed to provide a centralized Salesforce-based solution for managing event planning operations, including client bookings, event scheduling, venue reservations, vendor coordination, cancellation approvals, reminders and feedback collection. The system enables event teams to manage Events, Clients, Vendors, Venues and Feedback using a structured Salesforce data model with appropriate relationships.

Automated Flows handle event reminders and process updates, while an Approval Process manages event cancellation requests. Apex Classes and Triggers support advanced business logic such as venue availability updates and prevention of double bookings. A Lightning App, Reports and Dashboards provide visibility into event schedules, vendor activities and operational performance.

1. INTRODUCTION

1.1 Project Overview

EventForce is a Salesforce-based CRM solution developed to centralize and streamline event planning and management operations. It manages clients, events, venues, vendors, feedback and event cancellations within a structured Salesforce platform. The solution uses custom objects, relationships, validation rules, Flows, Approval Processes, Apex Triggers and Classes, Reports and Dashboards.

1.2 Purpose

- Centralize client, event, venue, vendor and feedback information.
- Streamline event booking and venue reservation processes.
- Prevent double booking through automated Apex logic.
- Automate event reminders and cancellation approval processes.
- Improve coordination between clients, coordinators, vendors and venue managers.
- Provide reports and dashboards for operational visibility and decision-making.
- Reduce manual processes, improve data accuracy and enhance client experience.

2. IDEATION PHASE

2.1 Problem Statement

PS	Customer	Problem / Need
PS-1	Event Operations Manager	Client, event, venue and vendor information is difficult to manage centrally.
PS-2	Event Coordinator	Event details and venue reservations require multiple manual activities.
PS-3	Venue Manager	Venue availability must be tracked to prevent double booking.
PS-4	Vendor Manager	Vendor details and event assignments need coordinated management.
PS-5	Event Coordinator	Clients need timely event information and reminders.
PS-6	Event Administrator	Cancellation requests require controlled approval and status updates.
PS-7	Event Coordinator	Feedback needs to be maintained with event and client details.
PS-8	Event Operations Manager	Operational information needs to be organized into reports and dashboards.

2.2 Empathy Map Canvas

User Persona: Event Coordinator

- Think & Feel: Wants smooth event management, avoids booking conflicts and provides reliable client service.
- Hear: Requirements and updates from clients, vendors, venue managers and administrators.
- See: Event details, client information, venue availability, vendor assignments, cancellations and feedback.
- Say & Do: Coordinates stakeholders, updates records, manages bookings and monitors event progress.
- Pain: Manual coordination, fragmented information, venue double-booking risk and difficulty tracking activities.
- Gain: Centralized information, automated reminders, accurate venue availability and improved visibility.

2.3 Brainstorming & Idea Prioritization

Idea	Approach	Key Limitation
Manual Registers	Maintain event, client, venue and vendor details manually	Time-consuming
Spreadsheet Tracking	Track bookings and event information using spreadsheets	Limited automation
Standalone Portal	Develop a dedicated event-management portal	Higher development and maintenance effort
Salesforce CRM	Centralize event operations on Salesforce	Requires proper configuration

Final Selection: Automation, Scalability and Real-Time Visibility. Salesforce CRM was selected because it provides the centralized, automated and scalable platform required for the project.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stage	Customer Journey Stage	What Happens
1	Event Enquiry & Client Registration	Client requirements and contact details are captured in Salesforce.

Stage	Customer Journey Stage	What Happens
2	Event Planning & Booking	Event type, date, budget and status are created and managed.
3	Venue Selection & Reservation	Suitable venues are identified, availability is checked and reservations are managed.
4	Vendor Coordination	Vendors and required services are assigned to events.
5	Event Confirmation & Communication	Confirmed events trigger automated reminders.
6	Event Execution & Monitoring	Progress is monitored through Salesforce records, reports and dashboards.
7	Cancellation & Approval	Cancellation requests follow the approval process and notifications.
8	Feedback & Closure	Client feedback is captured and completed events are updated.

3.2 Functional Requirements

FR	Requirement
FR-1	Event management
FR-2	Client management
FR-3	Venue management
FR-4	Vendor management
FR-5	Event–vendor assignment
FR-6	Venue availability & double-booking prevention
FR-7	Automated event reminders
FR-8	Event cancellation approval
FR-9	Feedback management
FR-10	Reports & dashboards
FR-11	Event status automation
FR-12	Data migration through Data Import Wizard

3.3 NON-FUNCTIONAL REQUIREMENTS

NFR	Description
Usability	Lightning App for easy event management and navigation
Security	Profile-, role- and permission-based access control
Reliability	Accurate automation, approvals and venue booking validation
Performance	Efficient processing of event and operational data
Availability	Cloud-based Salesforce access for authorized users
Scalability	Supports growing events, clients, vendors, venues and users
Maintainability	Flows, Apex, validation rules and security can be monitored and updated
Data Integrity	Validation and controlled migration maintain accurate records

3.4 DATA FLOW DIAGRAM

- Client / Event Team → Enter Client & Event Details → Salesforce EventForce System.
- Validate & Store Data → Event, Client, Venue & Vendor Records.
- Check Venue Availability → Assign Venue & Vendors → Confirm Event.
- Automated 3-Day Reminder → Event Execution & Status Updates → Feedback Collection.
- Reports & Dashboards → Event Operations & Decision Making.

Supporting Flows

- Cancellation Request: Event Coordinator → Cancellation Request → Approval Process → Approved / Rejected → Event Status Update → Notifications.
- Venue Booking Validation: Event → Apex Trigger → VenueStatusHelper → Check Existing Venue & Event Date → Prevent Double Booking / Update Venue Availability.
- Event Completion: Event Date Passed → Scheduled Apex → Batch Process → Event Status = Completed.

3.5 Technology Stack

Layer	Technology
Platform	Salesforce CRM / Developer Edition
UI	Salesforce Lightning App
Database	Salesforce Custom Objects
Business Logic	Salesforce Flows, Formula Fields, Validation Rules
Approval	Salesforce Approval Process

Layer	Technology
Automation	Record-Triggered Flows, Apex Triggers
Processing	Batch Apex, Scheduled Apex
Security	Profiles, Roles, Permission Sets, Sharing Rules, OWD
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Solution	Benefit
Information is difficult to manage centrally	Custom Event, Client, Venue, Vendor, Feedback and EventVendor objects	Centralized data
Booking involves manual activities	Salesforce Flows and automated processes	Reduced manual effort
Risk of venue double booking	Apex Trigger and VenueStatusHelper	Availability validation
Vendor coordination is difficult	EventVendor junction object	Event–vendor assignment
Clients may miss reminders	3-Day Reminder Flow	Automated communication
Cancellation needs control	Approval Process	Structured approval
Feedback needs event linkage	Feedback with Event and Client relationships	Organized feedback
Operational information is difficult to monitor	Reports and Operations Dashboard	Better visibility
Different users need different access	Profiles, Roles, Permission Sets, OWD and Sharing Rules	Controlled access

4.2 Proposed Solution

Parameter	Description
Problem	Manual and fragmented event management
Solution	Salesforce CRM
Innovation	Automated event reminders, venue validation and cancellation approvals
Impact	Faster event coordination and improved operational visibility
Business Model	Centralized event planning and management
Scalability	Supports growing events, clients, vendors, venues and users

4.3 SOLUTION ARCHITECTURE

- Event Object; Client Object; Vendor Object; Venue Object.
- EventVendor Junction Object; Feedback Object.
- Validation Rules; Formula Fields; Flows; Approval Process.
- Apex Triggers; Apex Classes; Batch Apex; Scheduled Apex.
- Profiles & Roles; Permission Sets & Sharing Rules.
- Data Import Wizard; Lightning App; Reports; Dashboards.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology.
- Epics → Stories → Story Points.
- Sprint-based execution.
- Velocity calculation used for estimation.

Product Backlog and Sprint Schedule

Sprint	Epic	User Story	Points	Priority
Sprint-1	Event Management	Create events with type, date and status	3	High
Sprint-1	Client & Venue	Associate client and venue with event	5	High
Sprint-2	Vendor Management	Assign multiple vendors to event	5	High
Sprint-2	Notifications	Provide timely event reminders	3	High
Sprint-3	Approval Management	Cancellation follows approval process	5	High
Sprint-3	Venue Management & Apex	Prevent double bookings	8	High
Sprint-4	Security	Role-based access	5	High
Sprint-4	Reports & Dashboards	Monitor event operations	5	Medium

6. PROJECT DEVELOPMENT PHASE

Phase 1: Requirement Analysis & Planning

The project was planned to help event planners manage client bookings, vendor coordination, venue reservations and feedback collection within a single CRM platform. The system automates scheduling, cancellation approvals and client reminders while providing visibility through reports and dashboards.

Functional Scope

- Custom objects: Event, Client, Vendor, Venue, Feedback and Event Vendor.
- Lookup, Master-Detail and Many-to-Many relationship modeling.
- Flows for reminders, feedback notifications and approval updates.
- Validation Rules and Formula Fields for accurate data entry and event budget calculation.
- Approval Processes for event cancellation requests with email alerts.
- Profiles, Roles and Permission Sets for access and security.
- Reports and Dashboards for event summary, vendor performance and upcoming-event insights.

Stakeholder Mapping

- Clients: Request event services, provide feedback and view updates.
- Event Coordinators: Manage scheduling, communication and approval workflows.
- Vendors: Handle assigned services such as catering, decor and logistics.
- Venue Managers: Manage venue availability and reservations.
- Event Administrators: Oversee operations, security and data monitoring.

Execution Roadmap

Object & Field Setup → Automation Flows → Validation Rules & Formula Fields → Profiles, Roles & Permission Sets → Approval Process → Apex Classes & Triggers → Reports & Dashboards.

Phase 2: Backend Development & Configurations

This phase built the core functional engine using Salesforce Flows, Apex logic and security configurations. Record-Triggered Flows, Validation Rules, Formula Fields and Approval Processes automated booking confirmations, reminders and cancellation approvals. Apex Classes and Triggers handled venue availability and double-booking prevention.

Milestone 1 – Salesforce Account

A Salesforce Developer Edition account was created as the development environment. User details, role, organization and a unique username were configured, followed by verification and password setup.

Milestone 2 – Objects

Salesforce objects act as database tables containing fields and records. EventForce uses Event, Client, Vendor, Venue, Feedback and EventVendor as its core objects.

Object	Fields / Purpose
Event__c	Event Name, Event Date, Event Type, Event Status, Event Budget
Client__c	Client Name, Email, Phone, Address, Country, City
Vendor__c	Vendor Name, Email, Phone, Service Type, Status
Venue__c	Venue Name, Address, Location URL, Capacity, Availability Status
Feedback__c	Feedback ID, Rating, Comments
EventVendor__c	Junction object linking Event and Vendor for Many-to-Many relationship

Relationships

Relationship	Type
Event → Client	Lookup
Event → Venue	Lookup
Feedback → Event	Lookup
Feedback → Client	Lookup
EventVendor → Event	Master-Detail
EventVendor → Vendor	Master-Detail

Milestone 3 – Tabs

Custom tabs were created for Event, Client, Vendor, Venue and Feedback so users can view object data through the Salesforce application.

Milestone 4 – Fields and Relationships

Event Type uses a Picklist containing Wedding, Corporate, Birthday, Anniversary, Festival, Concert and Other. Event Date uses a Date field, while Event Budget uses a Formula field based on Event Type. Client, Vendor, Venue and Feedback fields were created according to the application requirements.

Milestone 5 – Validation Rules

Validation Rules check whether entered data meets specified criteria before a record is saved. The Client__c rule Email_Valid_Address validates the email pattern and displays an error when an invalid address is entered.

Milestone 6 – Approval Process

An Event Cancellation Approval Process was implemented. A cancellation request uses the Pending Cancellation status, sends notification emails, assigns a manager for review and updates Event Status to Canceled after approval. Rejection actions notify the Event Owner.

Milestone 7 – Flows

The 3-Day Reminder Flow is a Record-Triggered Flow on Event. For confirmed events, a scheduled path runs three days before Event Date. Text templates prepare the reminder and a Send Email action sends it to the Client with the Event Owner copied.

Milestone 8 – Apex Classes

VenueStatusHelper updates Venue availability based on Event status. When an event is confirmed, the related venue can be marked Reserved; when an event is cancelled, it can be made Available according to the configured logic.

Milestone 9 – Apex Triggers

EventTrigger13 is an after-insert/after-update trigger on Event__c that invokes VenueStatusHelper. PreventDoubleBooking is a before-insert/before-update trigger that checks whether two events are assigned to the same venue on the same date.

- Collect venue IDs from new Event records.
- Query existing Event records for those venues.
- Compare venue and event-date combinations.
- Prevent conflicting records when a double booking is detected.

Milestone 10 – Batchable and Schedulable Asynchronous Apex

BatchCompleteEvents identifies past events whose status is not Completed and updates them to Completed. Scheduled Apex runs the batch job at a defined time each day, supporting automated event completion.

Phase 3: UI/UX Development & Customization

The Lightning App was configured with Event, Client, Vendor, Venue, Feedback, Reports and Dashboards. The EventForce Operations Dashboard provides a visual summary of key business metrics and supports data-driven monitoring.

Phase 4: Data Migration, Testing & Security

Validated records for Events, Clients, Vendors, Venues and Feedback were imported using the Data Import Wizard after pre-validation checks. User accounts were created for Event Admin, Event Coordinator, Vendor Manager and Client with appropriate Profiles and Roles.

Security configurations included Profiles, Role Hierarchy, Permission Sets and Sharing Rules. Organization-Wide Defaults were set to Private, with controlled access through sharing rules such as Event_Sharing_For_Vendors.

Security Components

- Profiles – baseline object, field, user, tab, app and Apex permissions.
- Roles – hierarchy and record visibility.
- Permission Sets – additional permissions without changing the base profile.
- Sharing Rules – controlled extension of record access.
- OWD – Private for controlled record access.

Reports

The Upcoming Events by Month report provides analytical tracking of upcoming events and counts. Event Date can be grouped by calendar month and Event Budget can be summarized.

Dashboards

The EventForce Operations Dashboard provides a visual summary of important business metrics and supports monitoring of schedules, vendor activities and feedback.

7. FUNCTIONAL & PERFORMANCE TESTING

Testing validated data accuracy, automation performance and user permissions. Configuration testing covered event creation, venue assignment, validation logic, scheduled processing, reminders and dashboard analytics.

Test Area	Expected Validation
Event Creation	Required event fields can be entered and saved.
Venue Assignment	Venue relationship and availability are handled correctly.
Double Booking	Conflicting venue/date combinations are prevented.
Reminder Flow	Qualifying confirmed events trigger the 3-day reminder.
Cancellation Approval	Request follows approval and notification steps.
Batch / Scheduled Apex	Past eligible events are updated to Completed.
Security	Access follows profiles, roles and sharing settings.
Reports & Dashboard	Operational data is displayed for monitoring.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized Data Management – Event, Client, Vendor, Venue, Feedback and EventVendor information is managed within Salesforce.
- Process Automation – Flows automate reminders and business processes, reducing manual effort.
- Prevents Double Booking – Apex Trigger and VenueStatusHelper validate venue availability.
- Controlled Cancellation Process – Approval Process provides structured handling of cancellation requests.
- Improved Operational Visibility – Reports and Dashboards provide visibility into schedules and activities.
- Secure Access – Profiles, Roles, Permission Sets, OWD and Sharing Rules provide controlled access.
- Scalable Solution – Salesforce supports future growth and additional business requirements.
- Better Data Quality – Validation Rules and pre-validation support accurate records.

Disadvantages

- Developer Edition Limitation – Implemented for academic purposes and not connected to production.
- Production Deployment Not Performed – Production migration was outside the scope.
- Deployment Requires Additional Tools – Real-world deployment may require sandboxes, Change Sets, Salesforce DX or DevOps/CI-CD.
- Configuration Complexity – Flows, Apex, approvals and security configurations require Salesforce knowledge.
- Ongoing Maintenance Required – Flows, Apex jobs, approvals, validation and sharing settings need monitoring.

9. CONCLUSION

The EventForce Management System demonstrates how Salesforce can be used to streamline and automate event planning and management operations. Events, Clients, Vendors, Venues, Feedback and EventVendor data are brought into a centralized platform.

The implementation combines Salesforce Lightning, Flows, Validation Rules, Formula Fields, Approval Processes, Apex Triggers, Apex Classes, Batch Apex, Scheduled Apex, security controls, Reports and Dashboards. Automation supports event reminders, cancellation approvals, venue availability management, double-booking prevention and event completion.

The project also demonstrates the importance of data migration, testing, security, documentation and maintenance. The solution was developed and tested in a Developer Edition Org; production deployment was not performed.

10. FUTURE SCOPE

- Production Deployment – Deploy to a production Salesforce environment using appropriate deployment tools.
- Advanced Event Automation – Extend automation for additional event planning and operational activities.
- Enhanced Vendor Management – Add advanced vendor tracking and performance management.
- Advanced Analytics – Add deeper reports and dashboards for events, venues, vendors and feedback.
- External Integrations – Integrate Salesforce with external systems and services.
- Mobile Accessibility – Improve event management through mobile devices.
- Scalability Enhancements – Support larger numbers of events, clients, vendors, venues and users.
- Continuous Maintenance & Optimization – Regularly improve Flows, Apex, validation and sharing settings.

11. APPENDIX

A. Custom Objects

- Event
- Client
- Vendor
- Venue
- EventVendor
- Feedback

B. Automation

- Flows
- Validation Rules
- Formula Fields
- Approval Process
- Apex Triggers & Classes

C. Processing

- Batch Apex
- Scheduled Apex

D. Security

- Profiles
- Roles
- Permission Sets
- OWD
- Sharing Rules

E. Data Migration

- Salesforce Data Import Wizard

F. UI & Analytics

- Lightning App
- Reports
- Dashboards

G. Testing & Documentation

- Configuration testing
- Security testing
- Automation testing
- Project documentation

THANK YOU